

RESEARCH ARTICLE | *Fluid and Electrolyte Homeostasis*

Tricellular tight junction-associated angulins in the gill epithelium of rainbow trout

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Kolosov D, Kelly SP. Tricellular tight junction-associated angulins in the gill epithelium of rainbow trout. *Am J Physiol Regul Integr Comp Physiol* 315: R312–R322, 2018. First published April 6, 2018; doi:10.1152/ajpregu.00431.2017.—Molecular physiology of the tricellular tight junction (tTJ)-associated proteins lipolysis-stimulated lipoprotein receptor (*Lsr*, = angulin-1) and an immunoglobulin-like domain-containing receptor (*ilDR2*, ≈ angulin-3) was examined in model trout gill epithelia. Transcripts encoding *Lsr* and *ilDR2* are broadly expressed in trout organs. A reduction in *Lsr* and *ilDR2* mRNA abundance was observed during and after confluence in flask-cultured gill cells. In contrast, as high-resistance and low-permeability characteristics developed in a model gill epithelium cultured on permeable polyethylene terephthalate membrane inserts, *Lsr* and *ilDR2* transcript abundance increased. However, as epithelia entered the developmental plateau phase, *Lsr* abundance returned to initial values, while *ilDR2* transcript abundance remained elevated. When mitochondrion-rich cells were introduced to model preparations, *Lsr* mRNA abundance was unaltered and *ilDR2* mRNA abundance significantly increased. Transcript abundance of *ilDR2* was not altered in association with corticosteroid-induced tightening of the gill epithelium, while *Lsr* mRNA abundance decreased. Transcriptional knockdown of the tTJ protein tricellulin (Tric) reduced Tric abundance, increased gill epithelium permeability, and increased *Lsr* without significantly altering *ilDR2* transcript abundance. Data suggest that angulins contribute to fish gill epithelium barrier properties but that *Lsr* and *ilDR2* seem likely to play different roles. This is because *ilDR2* typically exhibited increased abundance in association with decreased model permeability, while *Lsr* abundance changed in a manner that suggested a role in Tric recruitment to the tTJ.

angulin; cortisol; epithelium; gill; tricellulin

INTRODUCTION

The tight junction (TJ) complex of vertebrate epithelia acts as a paracellular diffusion barrier and comprises regions of bicellular contact between two epithelial cells [the bicellular TJ (bTJ)], as well as regions of tricellular contact (TC), where three epithelial cells interface [the tricellular TJ (tTJ)]. While knowledge of bTJ function has burgeoned since the discovery of integral bTJ proteins (13, 14), the tTJ and the function of its molecular components remain enigmatic (15). The tTJ complex was first reported as a structure that occurred at angular TC points of epithelial cells in the plane perpendicular to the epithelial sheet (33). However, identification of TJ proteins that associate with regions of TC did not occur until tricellulin

(TRIC) was first described as a tTJ-associated membrane protein of murine epithelia (19). TRIC was proposed to contribute to the maintenance of vertebrate epithelium integrity, because RNAi-mediated *Tric* knockdown (KD) in a model mammary epithelium led to a reduction in transepithelial electrical resistance (TER) and an increase in paracellular marker flux (19). Recently, additional proteins, which have been collectively designated “angulins” (17), have been reported as tTJ-associated elements. These include lipolysis-stimulated lipoprotein receptor (LSR, angulin-1) and immunoglobulin-like domain-containing receptors 1 (ILDR1, angulin-2) and 2 (ILDR2, angulin-3) (17, 31). Angulin-1/LSR was reported as a tTJ-associated protein that localizes at regions of epithelial cell TC in a manner very similar to TRIC (31). Furthermore, KD of angulin-1/*Lsr* in a mammary epithelium model not only reduced TER and increased paracellular tracer flux, but also impaired recruitment of TRIC to the tTJ (31). These observations support the idea that angulin-1/LSR may be important for barrier function in vertebrate epithelia. Furthermore, if angulin-2/ILDR1 or angulin-3/ILDR2 is introduced to cells following angulin-1/*Lsr* KD, their presence also allowed TRIC to localize correctly to regions of TC, although angulin-2/ILDR1 was more effective than angulin-3/ILDR2 (17). In epithelial tissue of nonmammalian vertebrates, a role for angulins in barrier integrity has not been examined.

The gill epithelium of freshwater (FW) fishes is a high-resistance tissue barrier that separates blood from surrounding water. Transcellular ion transport properties of the FW fish gill epithelium have been extensively examined (10, 32). In contrast, the molecular properties of the paracellular pathway are only now being revealed (7). Most studies on the molecular characteristics of the FW fish gill paracellular pathway have focused on the molecular physiology of bTJ proteins (7, 27). However, recent studies using a model FW fish gill epithelium have provided evidence that the tTJ protein Tric plays an important role in the maintenance of gill epithelium integrity (24). Tric localizes to regions of TC in the fish gill epithelium, and treatment of a model gill epithelium with cortisol increased TER, reduced paracellular flux of [³H]polyethylene glycol (4,000 Da, PEG 4000), and increased Tric abundance (24). This provided a link between reduced gill epithelium permeability and elevated levels of a vertebrate barrier-forming tTJ protein. In addition, if regions of TC in the model gill epithelium were disrupted using sodium caprate, Tric was displaced, TER decreased, and [³H]PEG 4000 permeability increased (24). Because the caprate-induced change in epithelium permeability occurred while bTJs appeared to remain intact, these data support the idea that TC integrity is a critical component

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of gill barrier function and that the tTJ protein Tric is likely involved (24). Given the close association between the tTJ-associated protein Tric and angulins in terrestrial vertebrates, it seems reasonable to consider the idea that TJ-associated angulins will also participate in maintenance of epithelial integrity of aquatic vertebrates. In this regard, it can be hypothesized that TJ-associated angulins will respond to factors that influence gill epithelium permeability. The goal of the current study was to identify angulins of rainbow trout and explore how these tTJ-associated factors respond to manipulations of trout gill epithelium permeability.

MATERIALS AND METHODS

Experimental animals. Rainbow trout (*Oncorhynchus mykiss*, Walbaum 1792; Humber Springs Trout Hatchery, Orangeville, ON, Canada) were held in flow-through dechlorinated FW (in μM : 590 Na^+ , 920 Cl^- , 760 Ca^{2+} , and 43 K^+ ; pH 7.35), exposed to a constant 12:12-h photoperiod, and fed ad libitum once daily using a commercial trout pellet (Martin Profishent, Elmira, ON, Canada). All experimental procedures and animal care were conducted under an approved York University Animal Care Committee protocol that conformed to the guidelines of the Canadian Council on Animal Care.

Identification of angulins and sampling for expression profile analysis. Rainbow trout *lsr* and *ildr2* sequences were identified using previously published approaches (29). Briefly, expressed sequence tags (ESTs) from the rainbow trout genome similar to *LSR* and *ILDR* sequences from mouse and zebrafish were sought using a National Center for Biotechnology Information database BLAST search. Newly found ESTs were confirmed to be protein-coding, and primers were designed for PCR amplification.

For sampling, fish were net-captured and anesthetized in buffered tricaine methanesulfonate (1.0 g/l; MS-222, Syndel Laboratories). Fish were then killed by spinal transection and placed on ice for dissection. Discrete organs, including brain, eye, gill, esophagus, pyloric ceca, anterior intestine, middle intestine, posterior intestine, kidney, and skin, were carefully dissected, placed in TRIzol reagent (Invitrogen Canada, Burlington, ON, Canada), and frozen in liquid nitrogen, and samples were stored at -80°C until further analyses.

Procedures for preparation and primary culture of gill epithelia. The primary-cultured gill epithelia used in these studies were prepared using methods originally developed and described in methodological detail elsewhere (11, 23, 36). Models were composed of gill pavement cells (PVCs) only or PVCs and mitochondria-rich cells (MRCs). For gill PVC models, trypsin is used to harvest gill cells, which are seeded into cell culture flasks (25 cm^2 ; BD Falcon, BD Biosciences, Mississauga, ON, Canada). Once seeded, cells attach to the flask bottom and develop into a confluent epithelium over the course of ~6 days. During the initial stage of flask culture, only gill PVCs attach, and all other cells are washed out during media changes. This facilitates formation of an epithelium composed of gill PVCs (only on a solid substratum) and provides a model that can be used to examine changes in the molecular physiology of gill PVCs during the formation of epithelium confluence. Flask-cultured PVCs are bathed in L15 medium supplemented with 6% fetal bovine serum (FBS). Thereafter, confluent flask-cultured PVCs can be retrypsinated, harvested, and seeded onto semipermeable polyethylene terephthalate filter inserts (0.9-cm^2 area, $0.4\text{-}\mu\text{m}$ pore, 1.6×10^6 pores/ cm^2 density; BD Falcon). Cell culture inserts are housed in companion cell culture plates (BD Falcon), and the apical and basolateral compartments of the culture system contain L15 medium supplemented with 6% FBS. Insert-cultured PVCs rapidly become confluent, and, over the course of 5–6 days, the confluent PVC epithelium model develops a high TER that plateaus upon maturity. The insert-cultured gill epithelium model composed of PVCs only is referred to as the single-seeded insert (SSI) preparation. Because the SSI preparation rapidly becomes confluent and, over

time, develops a high resistance (and low permeability), it can be used to examine changes in the molecular physiology of a gill epithelium (composed of PVCs only) during development of its barrier properties. To prepare a cultured gill epithelium composed of PVCs and MRCs, gill cells are harvested using trypsin, as described above, but the gill cells are directly seeded onto semipermeable polyethylene terephthalate filter inserts (see above; BD Falcon). Inserts are also housed in companion cell culture plates, with the apical and basolateral compartments of the culture system containing L15 medium supplemented with 6% FBS. However, 24 h after these cells are seeded, the process is repeated, and cells from a second fish are directly seeded onto the same semipermeable polyethylene terephthalate filter inserts, which appears to allow MRCs to attach to the filter inserts. Then the epithelium containing both PVCs and MRCs develops a high resistance and reaches a plateau after 5–7 days. The epithelium model composed of PVCs and MRCs is referred to as the double-seeded insert (DSI) preparation.

To confirm the presence of MRCs in DSI preparations (and the absence of MRCs in SSI preparations), Na^+/K^+ -ATPase (NKA) immunoreactivity (ir) was used as a marker, because MRCs exhibit intense NKA-ir compared with PVCs, which exhibit weak NKA-ir. In this regard, MRCs are easily identified as brightly fluorescent cells. Procedures for NKA localization were conducted in accordance with those outlined previously (29) using an α -subunit antibody ($\alpha 5$; Developmental Studies Hybridoma Bank, Iowa City, IA).

When cultured (SSI only) preparations were exposed to cortisol, the treatment protocol was in accordance with methods outlined previously (21). More specifically, a cortisol stock solution was prepared as follows: cell culture-grade hydrocortisone 21-hemisuccinate sodium salt (Sigma-Aldrich) was dissolved in phosphate-buffered saline (PBS, pH 7.7) and stored in aliquots at -30°C until use. When medium was changed, stock cortisol solution was added to yield a final medium cortisol concentration of 500 ng/ml. Cortisol was only added to medium bathing the basolateral side of the preparation, and the dose was chosen based on previous dose-response experiments (21) and because similar levels of circulating cortisol have been reported in stressed rainbow trout (1).

Measurement of TER and $[^3\text{H}]$ polyethylene glycol flux. TER was measured in SSI and DSI (i.e., insert-cultured) preparations only with use of chopstick electrodes (STX2, World Precision Instruments, Sarasota, FL) attached to a custom-modified EVOM epithelial voltohmmeter (World Precision Instruments). Measured TER values were background-corrected by subtraction of TER of a blank insert containing medium of appropriate composition. All TER measurements are expressed in $\Omega\text{-cm}^2$. For SSI and DSI epithelia, flux of $[^3\text{H}]$ polyethylene glycol (molecular mass 400 Da, $[^3\text{H}]$ PEG 400; American Radio Chemicals, St. Louis, MO) was measured in the basolateral-to-apical direction before cells were harvested. $[^3\text{H}]$ PEG 400 flux was determined according to methods and calculations reported previously (37).

Quantitative real-time PCR analysis of angulins and tric. Transcript abundance of *lsr*, *ildr2*, and *tric* was determined by quantitative real-time PCR (qPCR) using techniques described in detail elsewhere (24). Briefly, total RNA was isolated from discrete organs (for *lsr* and *ildr2* expression profiles) or cultured gill epithelia using TRIzol reagent (Invitrogen Canada) according to the manufacturer's instructions. Total RNA was quantified using a Multiskan Spectrum UV/Vis microplate spectrophotometer (Thermo Fisher Scientific, Nepean, ON, Canada); thereafter, $2\text{ }\mu\text{g}$ of total RNA were treated with DNase I (amplification grade; Invitrogen Canada) and used for cDNA synthesis with SuperScript III reverse transcriptase and oligo(dT)-(12–18) primers (Invitrogen Canada). Primers used to determine transcript abundance were as follows: *tric* [GenBank accession no. KC603902; GTCACATCCCCAAACCAGTC (forward) and GTCCAGCTCGTCAAACCTTCC (reverse); predicted amplicon size 170 bp], *lsr* [GenBank accession no. BK009921; ACCCCCAGCCACCCTAC (forward) and GGAAGTACCTCGCTCA (reverse); predicted amplicon

size 347 bp], and *ildr2* [GenBank accession no. BK009920; CGGAA-GACACTATCAGGAGACT (forward) and CAGGTTTGTGGGCAGCAG (reverse); predicted amplicon size 182 bp]. Transcript abundance was determined with SYBR Green I supermix (Bio-Rad Laboratories Canada, Mississauga, ON, Canada) in a Chromo4 detection system (model no. CFB-3240, Bio-Rad Laboratories Canada). Reaction conditions were as follows: 1 denaturation cycle (95°C for 4 min) followed by 40 cycles of denaturation (95°C for 30 s), annealing (*tric* at 61°C, *lsr* at 60°C, and *ildr2* at 57°C for 30 s), and extension (72°C for 30 s). A dissociation curve analysis was carried out after each qPCR analysis was run to confirm the synthesis of a single PCR product. Transcript abundance was normalized to that of rainbow trout elongation factor $\alpha 1a$ [*ef-1a*; GenBank accession no. AF498320.1; GGCAAGTCAACCACCACAG (forward) and GATACCACGCTCCCTCTCAG (reverse); predicted amplicon size 159 bp, annealing temperature 60°C]. Use of *ef-1a* for gene-of-interest normalization was validated by statistical comparison of *ef-1a* threshold cycles between treatments to confirm that experimental conditions had not significantly altered *ef-1a* mRNA levels.

Transcriptional KD of *tric*. Custom-made small interfering RNA (siRNA) spanning nucleotides 964–985 of *tric* transcript with a sequence of GCAGCUGUAGUCUACGUCATT/UGACGUAGACUACAGCUGCTT was ordered from Shanghai GenePharma (Shanghai, China), and 50 pmol were used for *tric* KD in SSI preparations. Lipofectamine 3000 (Invitrogen Canada) was used as a transfection reagent, and preparations were transfected 24 h after the first seeding onto inserts. Negative control scrambled siRNA (sequence: CGAGC-CGUCCUGAUUCUAATT/UUAGAAUCAGGACGGCUCCTT) preparations transfected using Lipofectamine 3000 were used as a control. For analysis of *tric* mRNA in transfected tissue, total RNA was isolated, cDNA was synthesized, and qPCR analysis was run as previously described (see *Quantitative real-time PCR analysis of angulins and tric*). To verify that *tric* KD translated into a loss of protein, Western blot analysis was used to examine Tric protein abundance in epithelial tissue.

Western blot analysis and Tric protein abundance. Western blot analysis of Tric was conducted using methods previously described (8). Briefly, insert-cultured cells were rinsed with ice-cold PBS (4°C) and then incubated in a lysis buffer (10 mM Tris-HCl, pH 7.5, 1 mM EDTA, 0.1 mM NaCl, and 1 mM PMSF) with 1:200 protease inhibitor cocktail (Sigma-Aldrich Canada). Epithelial tissue was then disrupted by repeated passage of cells in solution through a 26-gauge syringe needle, and “homogenate” was collected. Samples were centrifuged at 10,000 *g* for 10 min at 4°C, and supernatant was collected. The total protein concentration of the supernatant was determined using a Bradford protein assay (Sigma-Aldrich Canada). A total of 5 μ g of protein were used for Western blotting along with a 12% SDS-polyacrylamide gel. Wet transfer was performed to transfer gel-run protein samples from the gel to a polyvinylidene difluoride membrane at 100 V for 1 h. After transfer, the membrane was incubated with 5% skim milk solution for 1 h at room temperature, and the membrane was incubated (overnight at room temperature) with a rabbit anti-Tric custom-synthesized antibody. The custom-synthesized Tric polyclonal antibody was raised in rabbit against a synthetic peptide (Ac-PDPHLELDPALDIK-amide) corresponding to a 74- to 87-amino acid region of rainbow trout Tric (GenScript). A 1:1,000 dilution in Tris-buffered saline with Tween (10 mM Tris, 150 mM NaCl, and 0.05% Tween 20; pH 7.4) was used. Signal detection was performed by incubation of the membrane with a horseradish peroxidase-conjugated goat anti-rabbit secondary antibody (Bio-Rad Laboratories Canada) for 1 h at room temperature. Antigen reactivity was examined by incubation of the membrane with Clarity Western ECL blotting substrate (Bio-Rad Laboratories Canada) for 5 min at room temperature. Imaging of Tric-ir was conducted using a Chemi-Doc MP System (Bio-Rad Laboratories Canada). Specificity of the custom-made primary antibody was confirmed using a peptide preabsorption technique described in detail elsewhere (4).

For quantification of Tric protein abundance, Western blot analysis was used to image Tric-ir, as described above. Membranes were then incubated in stripping buffer, blocked with 5% skim milk solution as outlined above, and incubated with mouse monoclonal anti-actin JLA20 antibody (Developmental Studies Hybridoma Bank). Actin-ir was imaged and used for normalization of Tric abundance after validation, by statistical comparison, that actin protein abundance between experimental groups did not significantly change in response to experimental treatment. Tric and actin protein abundance was quantified using ImageJ software [National Institutes of Health, Java 1.6.0_45 (64-bit)].

Statistical analysis. Statistical analysis was determined by Student's *t*-test or one-way ANOVA coupled with a multiple-comparison test, where appropriate, using SigmaPlot (version 11) statistical software. Statistical significance was based on the observation of a fiducial limit of $P < 0.05$.

RESULTS

Angulin phylogeny and expression profiles. In this study genes encoding two angulin proteins in the rainbow trout have been identified. These two proteins cluster with vertebrate *LSR* (angulin-1) and *ILDR2* (angulin-3) (Fig. 1A). No *ildr1* (angulin-2) was identified in the present study. In rainbow trout, *lsr* was present in every organ examined except the kidney, and *lsr* was particularly abundant in the brain and eye (Fig. 1B). Rainbow trout *ildr2* was expressed in all organs examined in the current study. It was found to be comparatively abundant in the brain, eye, gill, skin, kidney, and middle intestine (Fig. 1C).

Angulin mRNA abundance during development of model gill epithelia. Cultured trout gill epithelia grown in cell culture flasks developed into a confluent monolayer of PVCs with clearly delineated cell borders over a ~6-day period, as visualized using phase-contrast microscopy (Fig. 2, A–C). During this period, transcriptional changes in *lsr* (Fig. 2D) and *ildr2* (Fig. 2E) took place. More specifically, *lsr* mRNA abundance declined over the first 120 h after the first cell seeding, and at 240 h *lsr* mRNA abundance remained low. However, between 120 and 240 h (i.e., at 192 h), *lsr* mRNA abundance was elevated and not different from values measured 24 h after the first seeding (Fig. 2D). Transcript abundance of *ildr2* also declined in flask-seeded gill cells during the early period of epithelium development (i.e., over the first 72 h) (Fig. 2E). Then, *ildr2* mRNA abundance increased and, by 240 h, exhibited levels greater than those at 72 h but modestly lower than those at 24 h after the first seeding (Fig. 2E).

During the development of SSI preparations, TER increased substantially, from ~70 $\Omega \cdot \text{cm}^2$ at 24 h to ~4,500 $\Omega \cdot \text{cm}^2$ at 108 h (Fig. 3A). Coupled with this was a marked and stepwise reduction in [^3H]PEG 400 permeability (Fig. 3B). TER data (and associated PEG data) in Fig. 3, A and B, represent the lag (24 h), log (72 h), and plateau (108 h) phases of SSI preparation development and are taken from previously published work (29). During the development of SSI preparations, *lsr* mRNA abundance exhibited a transient increase 48 h following cell seeding but otherwise remained unchanged (Fig. 3C). In contrast, *ildr2* mRNA abundance exhibited a significant increase during the early development period of SSI epithelia and plateaued during the later stage of epithelium development (Fig. 3D).

Angulin mRNA abundance in SSI and DSI model gill epithelia. Data comparing SSI and DSI model epithelia were generated using age-matched mature preparations that had

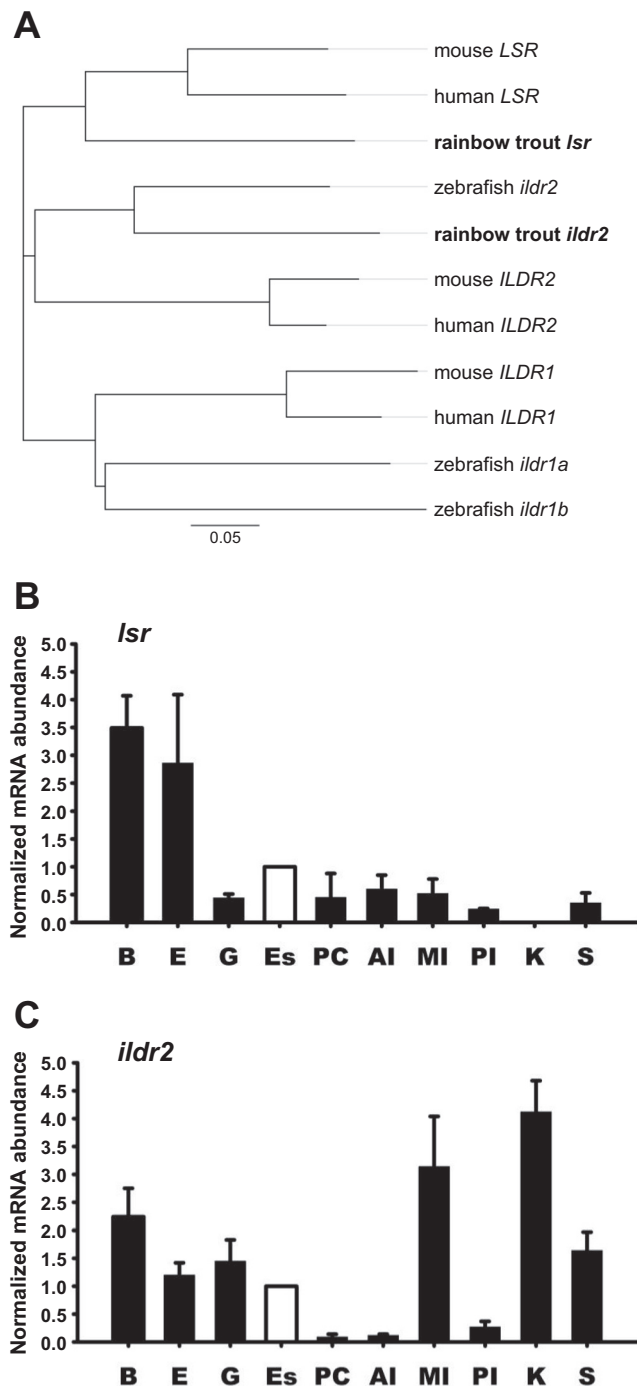


Fig. 1. Angulin phylogeny and expression profiles in discrete rainbow trout organs. **A**: a phylogenetic tree shows relatedness of human, mouse, zebrafish, and rainbow trout angulins. Note that the trout immunoglobulin-like domain-containing receptor gene (*ildr*) identified in this study clusters with zebrafish and mammalian *ildr2/ILDR2* and not the other two isoforms, allowing us to designate it trout *ildr2*. **B** and **C**: expression profiles of rainbow trout lipopolysaccharide-stimulated lipoprotein receptor gene (*lsr*) and *ildr2* in discrete organs show that these genes are broadly expressed, but also that mRNA abundance is not uniform. B, brain; E, eye; G, gill; Es, esophagus; PC, pyloric ceca; AI, anterior intestine; MI, middle intestine; PI, posterior intestine; K, kidney; S, skin. In **B** and **C**, fold change in mRNA abundance between organs is expressed relative to Es (white bar), which was assigned a value of 1.0. For *lsr* and *ildr2*, Es was selected as a reference, because it exhibited a quantitative PCR threshold cycle value of ~25. Values are means $\pm$ SE ($n = 4$).

developed on inserts for 108 h. TER was significantly greater in DSI than SSI preparations (Fig. 4A). Conversely, [3 H]PEG 400 permeability was significantly lower in DSI than SSI preparations (Fig. 4B). The presence of MRCs in the DSI preparations was confirmed using NKA-ir, NKA being an ion transport enzyme that is enriched in MRCs (Fig. 4C). In the DSI model epithelium, *lsr* mRNA abundance was not significantly different from that in the SSI model epithelium (Fig. 4D). However, *ildr2* transcript abundance was significantly elevated in DSI compared with SSI model epithelia (Fig. 4E).

Transcriptional response of angulins to corticosteroid treatment of a model gill epithelium. Treatment of a model gill epithelium (SSI preparations) with cortisol resulted in a significant increase in TER (Fig. 5A) and a significant reduction in [3 H]PEG 400 flux (Fig. 5B). In conjunction with these changes, *lsr* mRNA abundance decreased (Fig. 5C), while *ildr2* mRNA abundance was unaltered (Fig. 5D).

Effect of tric KD on Tric abundance, model gill epithelium permeability, and angulin transcript abundance. A custom-synthesized rainbow trout Tric antibody revealed a strong ~70-kDa immunoreactive band in gill epithelium tissue (Fig. 6A). Western blot analysis of Tric abundance in control and tric KD gill epithelium preparations demonstrated that tric KD significantly reduced Tric abundance (Fig. 6B). This was associated with a significant decrease in epithelium TER (Fig. 6C) and a significant increase in [3 H]PEG 400 flux (Fig. 6D). In addition, mRNA abundance of *lsr* increased in tric KD preparations (Fig. 6E), but transcript abundance of *ildr2* was not significantly altered (Fig. 6F).

DISCUSSION

Overview. In the current study two rainbow trout angulins (*lsr* and *ildr2*) are described. Expression profiles indicate widespread distribution of these genes. This is not unexpected, because evidence to date suggests that, in vertebrate epithelia, these tTJ-associated elements appear to contribute to the maintenance of epithelial integrity in regions of TC (15). The current study used cultured model gill epithelia, which are documented to closely mimic the passive electrical/transport characteristics of the intact gill and other surrogate gill models (11, 23, 36–38), to examine the transcriptional responses of *lsr* and *ildr2* during development, in the presence and absence of different cell types, and under conditions where epithelium permeability was manipulated by hormone treatment or KD of the integral tTJ protein Tric. Data provide evidence that Lsr and Ildr2 contribute to the maintenance of trout gill epithelium integrity, but the transcriptional response of *lsr* and *ildr2* appears to differ, to some extent, during development and, more so, in response to experimental manipulations that alter gill epithelium permeability. More specifically, alterations in transcript abundance of *ildr2* suggest that it may directly contribute to the establishment of resistive properties of cultured gill epithelia. In contrast, observations of *lsr* mRNA abundance indicate that this angulin may be involved in recruitment of Tric to regions of TC in a cultured gill epithelium, which would be consistent with its role in more-derived vertebrates (31).

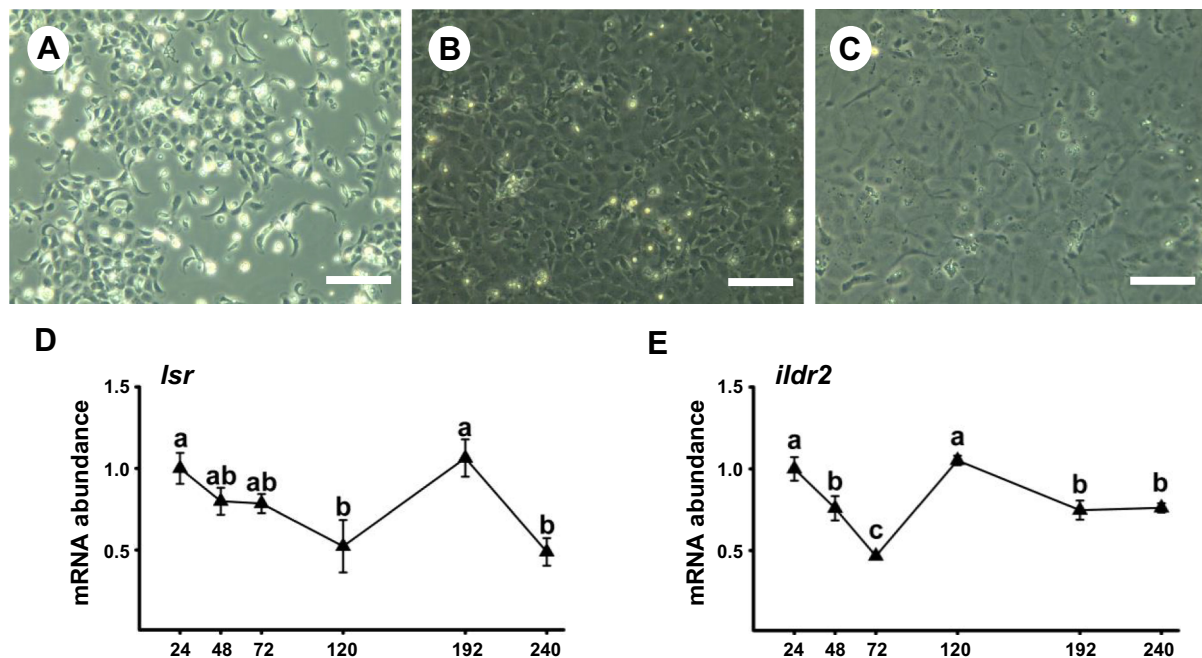


Fig. 2. Temporal responses of angulins during development of confluence in flask-cultured gill epithelial cells. A–C: images of cell confluence 24 (A), 72 (B), and 240 (C) h after gill cells were seeded into cell culture flasks. Scale bars = 150 μ m. D and E: temporal changes in transcript abundance of lipolysis-stimulated lipoprotein receptor (*lsr*) and immunoglobulin-like domain-containing receptor 2 (*ildr2*). Values are means $\pm$ SE ($n = 5$ –6). Different lowercase letters indicate significant differences between time points (by one-way ANOVA).

Expression of angulins in rainbow trout and alterations in transcript abundance during development of model gill epithelia. Angulins expressed by rainbow trout were similar to mammalian (and zebrafish) *LSR* and *ILDR2*. *ILDR* proteins are associated with the development of a robust barrier in murine epithelia, but, interestingly, *ILDR2* has been

associated with a weaker barrier function than *ILDR1* in mammals (17). However, mammalian *ILDR* isoforms demonstrate only ~30% amino acid sequence identity with each other and must have duplicated before the divergence of mammals from other vertebrates (17). In rainbow trout, *ildr2* does not appear to be associated with weak barrier

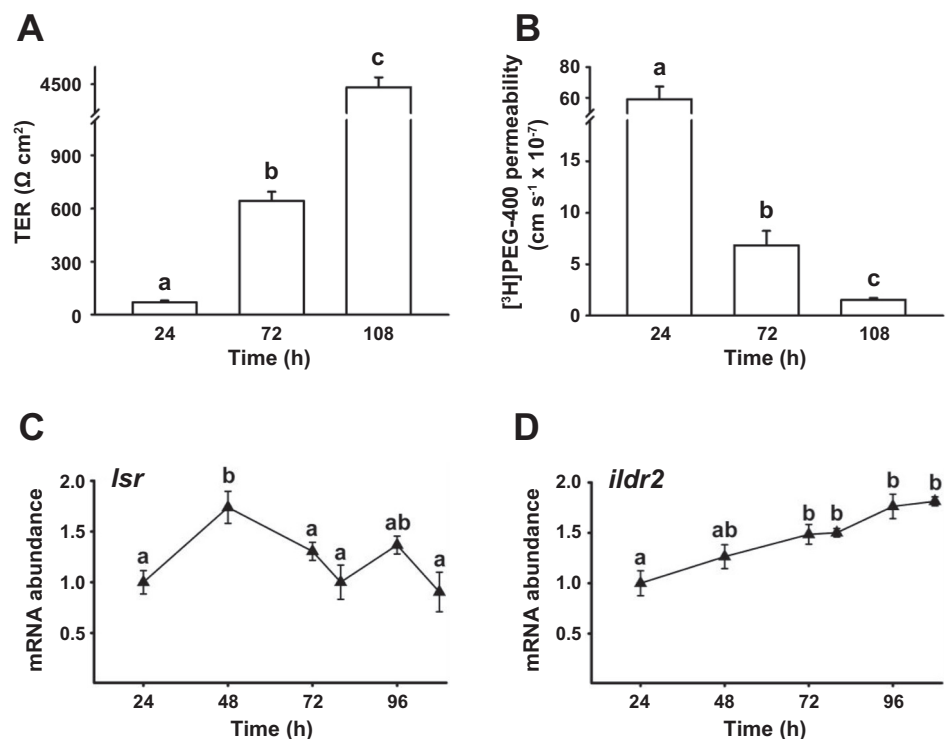


Fig. 3. Temporal responses of angulins during development of resistance in an insert-cultured model gill epithelium. A and B: alterations in transepithelial resistance (TER) and [³H]polyethylene glycol ([³H]PEG 400) flux at 24, 72, and 108 h during resistance development of an insert-cultured gill epithelium model composed of pavement cells (PVCs) only. Data in A and B are from Ref. 29. C and D: temporal changes in transcript abundance of lipolysis-stimulated lipoprotein receptor (*lsr*) and immunoglobulin-like domain-containing receptor 2 (*ildr2*) throughout development. Values are means $\pm$ SE ($n = 6$). Different lowercase letters indicate significant differences between time points (by one-way ANOVA).

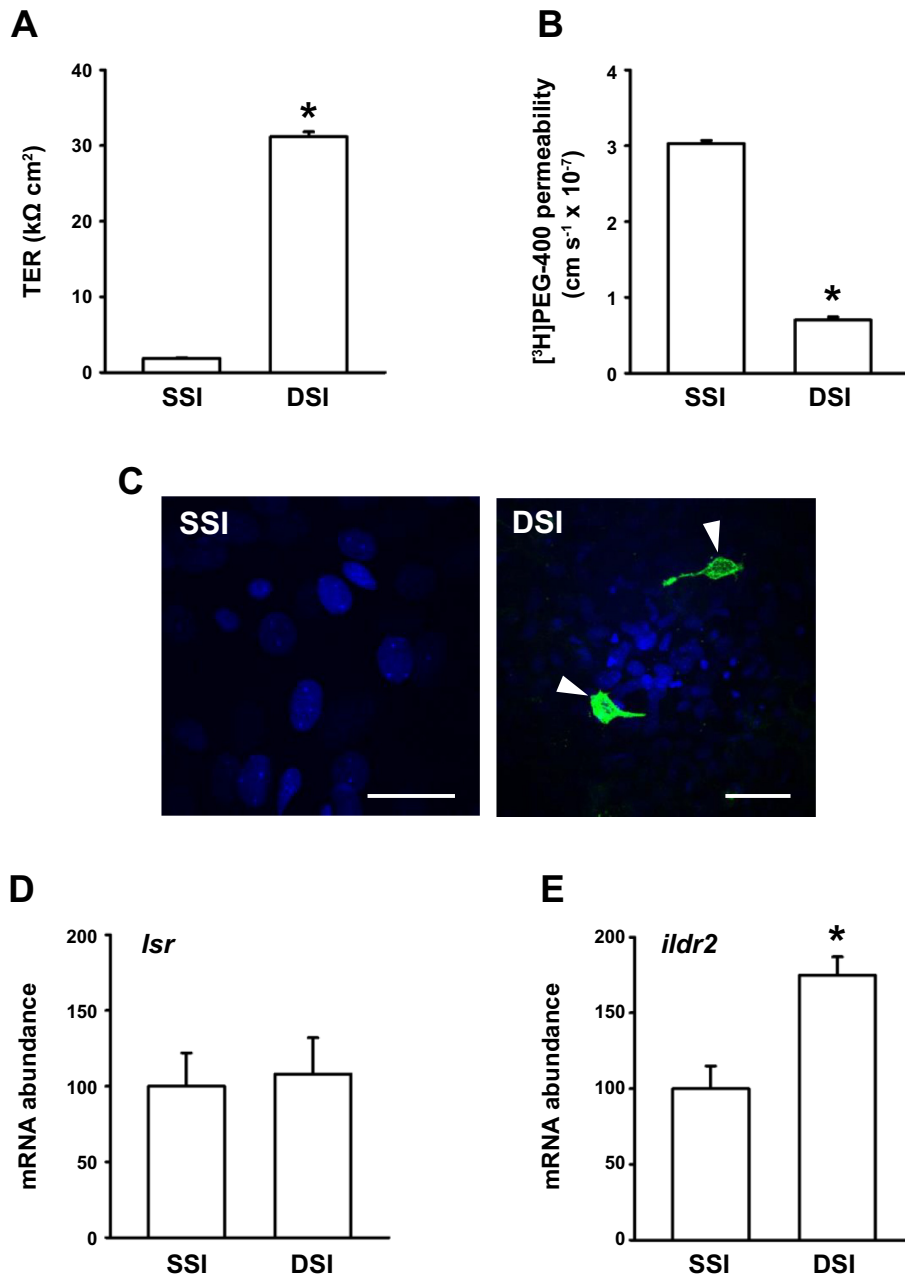


Fig. 4. Differences in angulin abundance between primary-cultured model gill epithelia that contain or do not contain mitochondrion-rich cells (MRCs). *A* and *B*: transepithelial resistance (TER) and [^3H]polyethylene glycol 400 ([^3H]PEG 400) flux across primary-cultured epithelial models that were composed of pavement cells (PVCs) only [single-seeded insert (SSI) preparations] or PVCs and mitochondrion-rich cells [MRCs; double-seeded insert (DSI) preparations]. *C*: $\text{Na}^+\text{-K}^+$ -ATPase immunoreactivity (green) used as a marker for detection of MRCs (white arrowheads). Scale bars = 25 μm (SSI) and 50 μm (DSI). Nuclei were stained with DAPI (blue). *D* and *E*: transcript abundance of lipolysis-stimulated lipoprotein receptor (*lsr*) and immunoglobulin-like domain-containing receptor 2 (*ildr2*) in SSI and DSI preparations. Values are means $\pm$ SE ($n = 10$). *Significantly different from SSI (by Student's *t*-test).

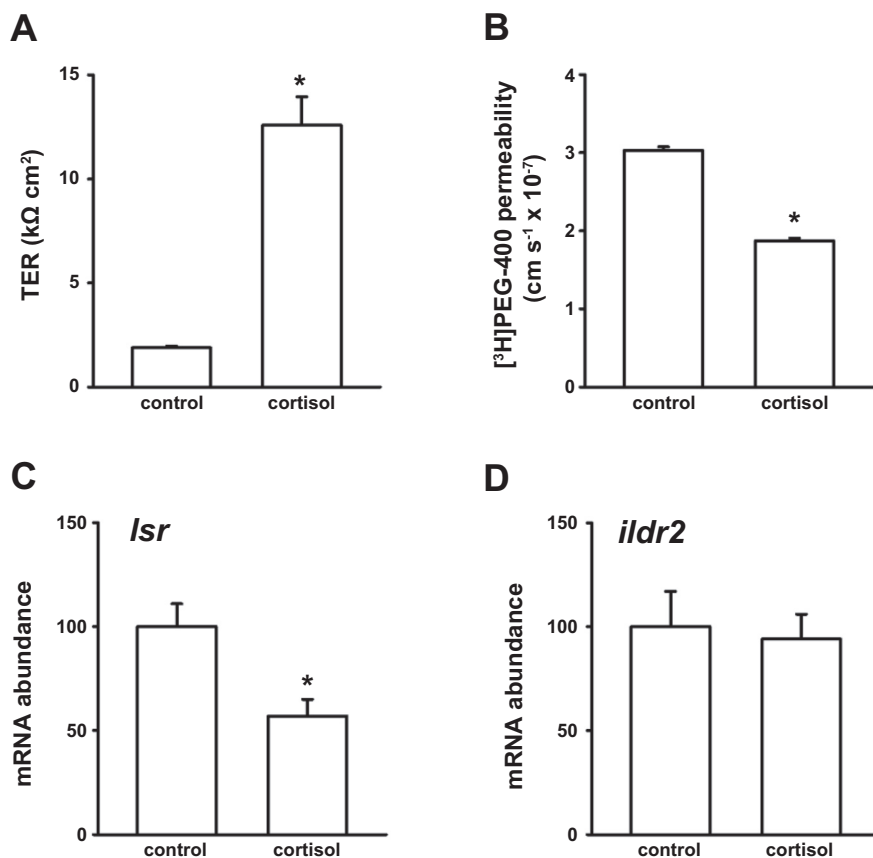
function, as the gill epithelium exhibits a very high resistance and, as such, is a tight barrier.

Both *lsr* and *ildr2* transcripts were broadly expressed in rainbow trout, which is consistent with the broad expression of genes encoding *tric*, an integral tTJ protein, in rainbow trout (24) and larval lamprey (28). However, in mammals, *LSR* and *ILDR* are often expressed in a complementary fashion, although several epithelia possess both (17). In rainbow trout, a curious observation was the absence of *lsr* mRNA in the kidney. The significance of this observation requires further examination, since *tric* is present in the kidney of trout, and in mammals *LSR* has been implicated in recruitment of TRIC to the tTJ (31). Nevertheless, *ildr2* transcript was particularly abundant in the trout kidney, and the complementary abundance of *ildr2* may, in part, compensate for the absence of *lsr*.

This would be consistent with previously observed complementary angulin expression profiles in mammals and is supported by evidence from mammalian studies suggesting that *ILDR* is also capable of recruiting TRIC to the tTJ (17, 31). In addition, *lsr* and *ildr2* transcripts were abundant in rainbow trout brain and eye. This suggests an important role for *lsr* and *ildr2* in barrier tissues of the trout brain and eye (e.g., blood-brain barrier, blood-retinal barrier, corneal epithelium, and conjunctival epithelium).

Despite some fluctuations, the mRNA abundance of *lsr* and *ildr2* was significantly lower at the end than the beginning of the flask-culture period. At the end of the flask-culture period, cells had formed a confluent monolayer with well-defined cell borders, indicative of TJ formation. In contrast, at the outset of the flask-culture period, gill cells were isolated as clusters or,

Fig. 5. Response of angulins to corticosteroid treatment of a model gill epithelium. *A–D*: effect of cortisol (500 ng/ml) on transepithelial resistance (TER), [^3H]polyethylene glycol ([^3H]PEG) 400 flux, lipolysis-stimulated lipoprotein receptor (*lsr*) mRNA abundance, and immunoglobulin-like domain-containing receptor 2 (*ilidr2*) mRNA abundance in a primary-cultured model gill epithelium composed of pavement cells only. Angulin transcript abundance was normalized to elongation factor $\alpha 1a$ (*ef-1a*) mRNA abundance and standardized to the control group (0 ng/ml cortisol added). Values are means $\pm$ SE ($n = 10$). *Significantly different from control (by Student's *t*-test).



to a lesser extent, as individuals. These observations suggest that the mRNA abundance of angulins in flask-cultured cells is greater at the outset to facilitate epithelium formation, but once an epithelium is formed and mature, angulins may take on a more restrained maintenance role.

Once flask-cultured cells were reseeded onto cell culture inserts, *lsr* and *ilidr2* exhibited a significant increase in mRNA abundance over the first 48–72 h. This occurs during the lag and early log phases of SSI development (29) and, as such, is consistent with the high levels of transcript abundance at the outset of flask culture. In this regard, both *Lsr* and *Ildr2* may play a similarly active role at the outset of development in both culture models. Somewhat in line with flask-culture observations, *lsr* mRNA abundance decreases (relative to the 48-h spike) during the later log and plateau phases of culture and, in this regard, may also take on a subtle maintenance role. However, *ilidr2* mRNA abundance remains high during the remaining log and plateau phases of SSI culture, suggesting that a more direct association between *Ildr2* and gill epithelium tightness may occur and that *Ildr2* might be a tTJ-associated element that contributes directly to the barrier properties of the trout gill epithelium.

Transcript abundance of angulins in model gill epithelia with and without MRCs. The development of different approaches toward the primary culture of gill epithelial cells has led to establishment of models that are composed of gill PVCs only (36) or both gill PVCs and gill MRCs (11). Under natural conditions, the majority of the gill-water barrier interface comprises the apical surface of gill PVCs. However, MRCs are also externally exposed and, in FW fishes, deep TJ adjoin

PVCs and MRCs, while the apical surfaces of MRCs are sites of intense ion transport activity (10). In primary-cultured model gill epithelia, TER is substantially higher in DSI than SSI preparations, and, in turn, permeability to paracellular permeability markers such as PEG is lower in DSI than SSI preparations (11). This means that when the molecular physiology of TJs in SSI preparations is compared with that in DSI preparations, any changes may reflect at least two factors: 1) the difference between SSI and DSI permeability and 2) the difference between cellular homogeneity (i.e., SSI composed of PVCs only) and cellular heterogeneity (DSI composed of PVCs and MRCs). Indeed, these influences may not be mutually exclusive, because TER has been positively correlated with MRC numbers in DSI preparations, suggesting that the TJs that link PVCs to MRCs in DSI epithelia may be “tighter” than those that link PVCs to PVCs (11). Whether this is actually the case (either in vitro or in vivo) is unknown, as we are unaware of any study that has directly examined TER across TJs that link FW fish gill MRCs to PVCs and compared values with those found across TJs between FW fish gill PVCs only. However, morphological evidence from the FW goldfish gill does not suggest that TJs that link PVCs to MRCs are tighter than those that link PVCs, because the linear depth of TJs adjoining FW goldfish gill MRCs and PVCs is not significantly different from that of TJs that link PVCs (6). In addition, when goldfish were acclimated from FW to ion-poor conditions (i.e., to an environmental condition that requires a further reduction in the paracellular permeability of the gill epithelium), TJs that linked gill PVCs as well as those that linked MRCs to PVCs increased, but more so between PVCs,

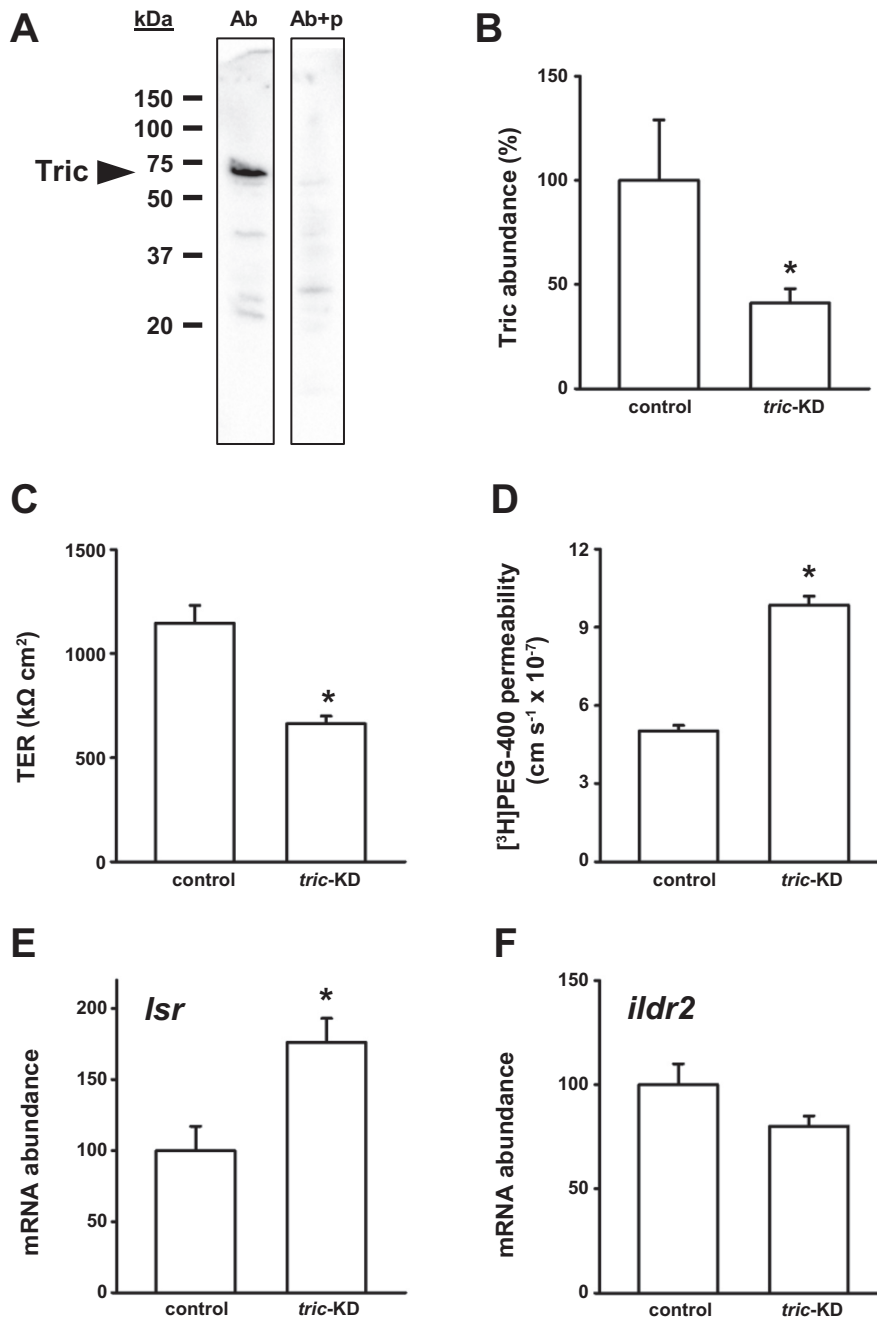


Fig. 6. Tricellulin (Tric) abundance, gill epithelium barrier properties, and response of angulins to transcriptional knockdown (KD) of *tric*. **A** and **B**: immunodetection of Tric and effect of *tric* KD on Tric abundance in a primary-cultured gill epithelium composed of gill pavement cells only. Ab, antibody; Ab + p, antibody + peptide (for preabsorption). **C–F**: effects of *tric* KD on transepithelial resistance (TER), [³H]polyethylene glycol ([³H]PEG) 400 flux, lipolysis-stimulated lipoprotein receptor (*lsr*) transcript abundance, and immunoglobulin-like domain-containing receptor 2 (*ildr2*) transcript abundance. Angulin transcript abundance was normalized to elongation factor $\alpha 1a$ (*ef-1a*) mRNA abundance and standardized to the control group. Values are means $\pm$ SE ($n = 6$). *Significantly different from control (by Student's *t*-test).

which resulted in significantly deeper TJs between gill PVCs versus gill MRCs-PVCs (6). In contrast, molecular differences between cultured gill epithelium models that arise from cellular heterogeneity versus cellular homogeneity are clearer cut. For example, a previous study compared the molecular components of DSI and SSI TJs and found that genes encoding select proteins (e.g., *cldn-10c* and *-10d*) are present in the DSI model but absent from the SSI model (29). Therefore, these genes are likely MRC-specific in the trout gill epithelium. In contrast, it was found that other genes exhibited greater, reduced, or unaltered abundance in DSI vs. SSI models.

In the case of *tric*, no difference between DSI and SSI models was previously observed in cultured trout gill epithelia (29), which is consistent with observations of *lsr* in the current

study. This suggests that *lsr* is not enriched in MRCs and that *lsr* abundance did not increase in accord with increased resistance in DSI preparations. Given that the development of a high resistance in SSI models was not linked to a significant increase in *lsr* transcript abundance, unaltered *lsr* mRNA abundance between DSI and SSI preparations would make sense. In contrast, *ildr2* exhibited greater mRNA abundance in DSI than SSI preparations. This suggests that *ildr2* is enriched in MRCs or that greater *ildr2* abundance reflects the higher resistance (lower permeability) of DSI preparations. Because *ildr2* transcript abundance significantly increased over time as SSI preparations developed a high resistance, it seems likely that higher DSI *ildr2* mRNA abundance can be at least partly related to the high DSI resistance. The knowledge that ILDR is

associated with the establishment of strong barrier properties in murine epithelia (17) supports this idea. The physiological significance of *ildr2*/ILDR2 contributing to enhanced barrier properties of the gill epithelium in the intact gill is that it may modulate the tTJ complex under conditions where gill barrier properties need to be augmented (e.g., upon ion-poor water exposure). For example, recent studies report that *tric* abundance is not elevated in the gill of rainbow trout following exposure to ion-poor conditions, which is in contrast to genes encoding proteins of the bTJ complex, many of which change significantly (9, 25). This is also in contrast to lamprey gills, where *tric*/Tric abundance was significantly elevated following exposure of animals to ion-poor water (28). In this regard, further study will be able to address the possibility that ILDR2 may be a tTJ protein of trout that enhances tTJ complex integrity in challenging environmental conditions. Moreover, whether *ildr2* is enriched in MRCs requires further investigation, with a logical approach being examination of *ildr2*/ILDR2 abundance in MRCs and PVCs isolated from the intact gill. If *ildr2*/ILDR2 is enriched in gill epithelium MRCs, it would be the first tTJ protein to exhibit this trend.

Effect of corticosteroid treatment on transcript abundance of angulins in a primary-cultured gill epithelium. Cortisol treatment of the gill epithelium has proved to be a useful tool for unraveling the mechanistic properties of the TJ complex and its protein composite in fishes (5, 8, 20, 26, 30). This is because cortisol treatment of primary-cultured gill epithelia significantly reduces the permeability of these preparations, and this occurs in association with reduced flux rates of paracellular permeability markers (5, 21, 22). In the current study cortisol elicited a response consistent with previous observations (21) i.e. a significant elevation of TER and significant reduction of [³H]PEG 400 flux. This would indicate that cortisol is tightening the gill epithelium model and that this occurs, at least in part, through a reduction in paracellular permeability. In association with these changes, *lsr* mRNA abundance decreased while *ildr2* mRNA abundance was unchanged. Consistent with the response of *lsr*, transcript abundance of the tTJ protein *tric* has been reported to decrease following cortisol treatment of a cultured gill epithelium model (24). However, a cortisol-induced reduction in *tric* abundance was observed in association with elevated Tric protein abundance (24). In a pancreatic cancer cell line, protein and mRNA abundance of TRIC has also been reported to uncouple following acute EGTA treatment (34), and this was attributable to TRIC phosphorylation. In the case of the former, it was suggested that this divergent response might occur as a result of extensively studied mRNA-regulatory mechanisms (2, 12), and it could be speculated that the same might apply to *lsr*/*Lsr* following cortisol treatment. In this regard, a cortisol-induced buildup of Tric protein in the gill epithelium was suggested to act as a negative feedback, resulting in a decrease in *tric* mRNA abundance (24). If *Lsr* is responsible for Tric recruitment to the tTJ complex in the fish gill epithelium (as it is in mammals), a reduction in *lsr* abundance, as observed in the current study following cortisol treatment, might also be expected. It will be essential to examine *Lsr* protein abundance to consider these possibilities further.

In contrast, cortisol treatment of a cultured gill epithelium did not significantly alter *ildr2* mRNA abundance. This observation would indicate that even though *ildr2* transcript abun-

dance was elevated under conditions where a high TER was developing or had developed in gill epithelium models (Figs. 3 and 4), a corticosteroid-induced increase in TER (reduction in permeability) may not be a gill barrier enhancement that involves elevated *ildr2*/ILDR2. In this regard, it is noteworthy that when *ildr2* mRNA abundance was elevated in association with increased gill epithelium resistance, such as during the development of SSI preparations (Fig. 3) or when DSI and SSI preparations are compared (Fig. 4), there is no evidence to suggest that *tric* abundance significantly changes (29). In contrast, cortisol treatment of the gill epithelium has been reported to increase Tric abundance (24), and in this study we see no response of *ildr2* following cortisol treatment. The physiological significance of this could be that both TRIC and ILDR2 play a role in enhancing the barrier properties of the tTJ complex in the fish gill epithelium but that they do so under different conditions and/or their response is mediated by different factors so as to avoid functional redundancy. However, these ideas are quite speculative and require further study with an emphasis on protein dynamics.

Effect of *tric* KD on gill epithelium barrier properties and transcript abundance of angulins. After *tric* KD, a significant reduction in Tric abundance was observed in association with a significant reduction in TER and increase in [³H]PEG 400 permeability. These observations are consistent with those of Ikenouchi et al. (19), who reported a significant reduction in TER and an increase in paracellular marker flux of Eph4 mammary epithelial cells in association with RNAi-mediated suppression of *Tric*. The response of the cultured gill epithelium to *tric* KD directly supports the idea that Tric is an important barrier protein in fishes, as previously suggested (24), and the idea that, in FW fish gill epithelium, Tric will help mitigate solute loss from extracellular fluid to the surrounding medium. These observations also underscore the importance of the tTJ complex in the regulated movement of solutes across epithelia of fishes and, in particular, the gill epithelium.

In association with *tric* KD, an increase in *lsr* mRNA abundance was observed in the primary-cultured gill epithelium. Because LSR has been reported to define regions of TC and recruit TRIC to the tTJ complex in mammals (16), it may be that increased *lsr* abundance in *tric* KD gill epithelium preparations reflected an attempt to increase Tric recruitment to regions of TC in a compromised barrier. A modest, but non-significant, decrease in *ildr2* abundance was also observed in *tric* KD preparations, which likely reflects the loss of barrier function.

Perspectives and Significance

The tTJ is an enigmatic structure, and little is known about the molecular structure and function of the tTJ compared with the bTJ. Studies conducted to date strongly support the idea that the tTJ and its molecular components play a significant role in maintaining the barrier properties of vertebrate epithelia (15). The current study provides new insight into the molecular physiology of the tTJ complex of a nonmammalian vertebrate. First, we considered how tTJ-associated elements (*lsr* and *ildr2*) respond to experimental manipulations associated with the formation and modulation of gill epithelium barrier properties. Second, we consolidated the idea that the tTJ protein Tric contributes significantly to barrier function in the

gill epithelium of fishes. In this regard, data suggest that continuing to uncover tTJ elements (e.g., angulins) and the complexities of the tTJ complex in fishes will be a valuable undertaking, in particular because we know from the current study and others (18, 24, 28, 35) that elements of the tTJ complex are responsive to developmental and environmental change, as well as diet and endocrine factors that mediate systemic change in response to these factors. In addition, the gene encoding Tric in Atlantic killifish has recently been suggested to contribute to adaptive divergence in the osmoregulatory physiology of this species (3), which further broadens the notion that the tTJ is important in the regulation of aquatic vertebrate salt and water balance.

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

D.K. and S.P.K. conceived and designed research; D.K. performed experiments; D.K. analyzed data; D.K. and S.P.K. interpreted results of experiments; D.K. and S.P.K. prepared figures; D.K. and S.P.K. drafted manuscript; D.K. and S.P.K. edited and revised manuscript; D.K. and S.P.K. approved final version of manuscript.

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